

Column Generation in Boolean Optimization

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Abstract. Recently several MaxSat algorithms based on SAT reduction have appeared in the literature. Our paper attempts to present a general methodology , gathering all the algorithms in a common framework.

1 Introduction

TO BE DONE: Max-2-SAT dual = graph duality. MaxSat dual is MinOnes? One can get it applying resolution. Maxflow, Mincut theorem?

In operations research, there are a well known set of column generation methods (e.g., Dantzig-Wolfe [4], Benders cuts [3]) which are based on the relaxation of an optimization problem into an easier problem (usually a linear problem solved by the Simplex method) and iteratively solve the initial problem.

Recently, a set of algorithms have been created to solve the MaxSat problem. Such algorithms are original in the sense that they iteratively reduce the MaxSat problem into Sat.

The complexity of MaxSat is $P^{NP[\log n]}$, meaning that it can be solved with a logarithmic number of calls to a SAT oracle [9].

In this paper we extend the idea of inference dual [7] to the MaxSat problem.

2 Preliminaries

Given a CNF instance ϕ , the MaxSat problem is the problem of finding an assignment which maximizes the number of satisfied clauses in ϕ .

The MaxSat problem can be written as

$$\min \sum s_i \tag{1}$$

$$\text{subject to } s_i + \sum a_{ij}x_i \geq 1 \tag{2}$$

$$a_{ij}, x_i, s_i \in \{0, 1\} \tag{3}$$

where x_i is a Boolean variable in the CNF and a_{ij} is the selector variable for variable x_i in clause j and s_j is the selector of clause j . Minimize the number of selectors in order to make the formula satisfiable.

The dual of the problem above is:

$$\min \sum \sigma_i \tag{4}$$

$$\text{subject to } \sigma_i + \prod \alpha_{ij}c_i = 1 \tag{5}$$

$$\alpha_{ij}, \sigma_i, c_i \in \{0, 1\} \tag{6}$$

$$\min \sum \sigma_i \quad (7)$$

$$\text{subject to } \sigma_i + F(\alpha_{ij}c_i) \leq 1 \quad (8)$$

$$\alpha_{ij}, \sigma_i, c_i \in \{0, 1\} \quad (9)$$

$$\max \sum \sigma_i \quad (10)$$

$$\text{subject to } \sigma_i + F(\alpha_{ij}c_i) \leq 1 \quad (11)$$

$$\alpha_{ij}, \sigma, c_i, F(\alpha_{ij}c_i) \in \{0, 1\} \quad (12)$$

where σ_i is the selector of the proof i and α_{ij} is the selector of clause i in proof j . Minimize the number of proofs with condition ≤ 1 in order to make the formula satisfiable. $F(\phi)$ is a function which returns 0 if the formula ϕ is satisfiable, otherwise returns 1.

This problem corresponds to

$$\max \sum s_i \quad (13)$$

$$\text{subject to } \prod s_i c_i = 0 \quad (14)$$

The minimal removal ($s_i = 0$) of c_i which makes the formula satisfiable.

An optimization problem can be written

$$\min\{f(x)|\mathcal{C}\} \quad (15)$$

where $f(x)$ is a real-valued function, $\mathcal{C}$ is a constraint set containing variables $x = (x_1, \dots, x_n)$, and D is a domain of x . A solution $\bar{x} \in D$ is feasible when it satisfies $\mathcal{C}$ and is optimal when $f(\bar{x}) \leq f(x)$ for all feasible x .

Given a CNF formula, the MaxSat is the problem of finding an assignment with the maximum number of satisfied clauses. Using the notation in 15, the MaxSat problem can be written as

$$\min_{a,x,w \in \{0,1\}} \{\sum w_j | \sum a_{ij}x_i + w_j \geq 1\} \quad (16)$$

The inference dual of 15 is the problem of finding the greatest lower bound on $f(x)$ that can be inferred from $\mathcal{C}$ within a given proof system. The inference dual can be written

$$\max_{P \in \mathcal{P}} \{v \mid \mathcal{C} \stackrel{P}{\vdash} (f(x) \geq v)\} \quad (17)$$

where $\mathcal{C} \stackrel{P}{\vdash} (f(x) \geq v)$ indicates that proof P deduces $f(x) \geq v$ from $\mathcal{C}$. The domain of variable P is a family $\mathcal{P}$ of proofs. A pair $(\bar{v}, \bar{P})$ is a feasible solution of 17 if $P \in \mathcal{P}$ and $\mathcal{C} \stackrel{\bar{P}}{\vdash} (f(x) \geq \bar{v})$, and it is optimal if $\bar{v} \geq v$ for all feasible (v, P) .

Equivalently, it can be defined as:

$$\min_{P \in \mathcal{P}} \{v \mid \mathcal{C} \stackrel{P}{\vdash} (f(x) \leq v)\} \quad (18)$$

Lemma 1. *Given a MaxSat primal assignment, there is a unique MaxSat dual assignment.*

Proof. Hitting set, unuseful variables, variables always in MUSes, ... [8]

2.1 Models for Column Generation

Barnhart et al. [2] considere a general problem P of the form:

$$\begin{aligned} \max \quad & c(x) \\ & Ax \leq b, \\ & x \in S, \\ & x \text{ integer} \end{aligned}$$

The function $c(x)$ is needed to be easy to solve with $Ax \leq b$ removed, and $S^* \subseteq S$. Its column generation form is:

$$\begin{aligned} \max \quad & \sum c_k \lambda_k \\ & \sum a_k \lambda_k \leq b \\ & \sum \lambda_k = 1 \\ & \lambda_k \in \{0, 1\} \quad k = 1, \dots, p \end{aligned}$$

3 Related work

Column generation in Integer Programming [2, 10]

Column generation in MaxSat [6]

Column generation in Constraint Programming [5]

4 Column generation in MaxSat

Application of the Dantzig-Wolfe decomposition to a structured optimization problem in which formulations with an exponential number of variables are obtained. Generating these variables on demand by means of delayed column generation is identical to performing a cutting plane on the respective dual problem.

Definition 1 (MUS). *A MUS ϕ is a CNF minimally unsatisfiable.*

Definition 2 (Relaxed CNF). *A relaxed CNF ϕ^r is a CNF ϕ with an additional (relaxing) variable for every clause, i.e.:*

$$\phi^r = \{c_i \vee b_i : \forall c_i \in \phi\}.$$

Notice that a relaxed CNF without any constraint to the variables b_i is satisfiable.

Definition 3 (Constrained Relaxed CNF). *A constrained relaxed CNF $\phi^r(k)$ is a relaxed CNF ϕ^r with an additional constraint over the relaxing variables $\sum r_i = k$ where $k \in \mathbb{N}$.*

Lemma 2. *Let $\phi^r(k)$ be a constrained relaxed CNF, ϕ be a CNF, and $f(\phi)$ be the number of falsified clauses in a MaxSat solution of ϕ . Then,*

$$f(\phi^r(k)) = \max(f(\phi) - k, 0).$$

Proof. A MaxSat solution of $\phi^r \cup (\sum b_i = k)$ has k different variables $b_i = 1$. Every of such variables correspond to a falsified clause in a MaxSat solution of ϕ , since this will bring the largest number of satisfied clauses. If $k \geq f(\phi)$, all the falsified clauses are assigned with $b_i = 1$, so $\phi^r \cup (\sum b_i = k)$ is satisfiable.

Notice that there is no need to relax the whole formula ϕ . Any falsified clause in a MaxSat solution belongs to a MUS. The set of clauses to be relaxed can be reduced to the set of clauses of k MUSes.

Given $\phi_1^r \cup (\sum b_i = k)$, ϕ_1^r relaxing the whole formula, and $\phi_2^r \cup (\sum b_i = k)$, ϕ_2^r relaxing k MUSes which contain a clause with $b_i = 1$, then both are MaxSat equivalent.

Corollary 1. *Given ϕ a CNF, $k \in \mathbb{N}$, and ϕ^r a relaxed CNF with $\sum b_i = k$, let k_m be the minimum k that makes the formula satisfiable, and let $f(\phi)$ be the number of falsified clauses in a MaxSat solution of ϕ . Then, $k_m = f(\phi)$.*

Proof. For any $k < f(\phi)$, there exists an falsified clause in any MaxSat solution for ϕ^r with $b_i = 0$, which makes ϕ^r unsatisfiable. For any $k \geq f(\phi)$, there exists at least one combination of falsified clauses in any MaxSat solution with $b_i = 1$ which makes the formula ϕ^r satisfiable.

Notice that, given a MUS, $k_m = 1$.

Lemma 3 (Composition of $\phi^r(k)$ is (sub)additive). *Given ϕ a CNF, ϕ_1^r a relaxed CNF ϕ with $\sum b_i = k_1 + k_2$, and ϕ_2^r a relaxed CNF ϕ with $\sum b_i^1 = k_1 \cup \sum b_i^2 = k_2$. Then ϕ_1^r and ϕ_2^r are MaxSat equivalent.*

Proof. Let $f(\phi)$ the number of falsified clauses in a MaxSat solution of ϕ . Assuming $k_1 + k_2 < f(\phi)$, then $f(\phi_1^r) = f(\phi) - (k_1 + k_2)$ and $f(\phi^r \cup \sum b_i^1 = k_1) = f(\phi) - k_1$. Adding constraint $\sum b_i^2 = k_2$ to formula $\phi^r \cup \sum b_i^1 = k_1$ reduces the number of unsatisfied clauses by k_2 , so $f(\phi^r \cup \sum b_i^1 = k_1 \cup \sum b_i^2 = k_2) = f(\phi) - k_1 - k_2$.

Corollary 2. *Given ϕ a CNF, and $\phi^{r(k)}$ a relaxed CNF with $\sum b_i^1 = 1 \cup \dots \cup \sum b_i^k = 1$. Let k_m be the minimum k that makes the formula $\phi^{r(k)}$ satisfiable, and let $f(\phi)$ be the number of falsified clauses in a MaxSat solution of ϕ . Then, $k_m = f(\phi)$.*

Proof. For any $k < f(\phi)$, there exists a set of falsified clauses in a MaxSat solution of ϕ^r with $b_i = 0$, which makes ϕ^r unsatisfiable. For any $k \geq f(\phi)$, there exists at least one combination of falsified clauses in a MaxSat solution of ϕ with $b_i = 1$, which makes the formula $\phi^{r(k)}$ satisfiable.

5 Integer Programming

$$\begin{aligned} \max \sum x_{ij} \\ \sum a_{ij} + x_{ij} \leq 1 \\ \sum x_{ij} = 1 \\ a_{ij}, x_{ij} \in \{0, 1\} \end{aligned}$$

Relaxation

$$\begin{aligned} \sum a_{ij} + x_{ij} \leq 1 \\ \sum x_{ij} = 1 \\ a_{ij}, x_{ij} \in \{0, 1\} \end{aligned}$$

Column generation

$$\begin{aligned} \sum a_{ij} + x_{ij} + y_{ij} \leq 1 \\ \sum x_{ij} = 1 \\ \sum y_{ij} = 1 \\ a_{ij}, x_{ij} \in \{0, 1\} \end{aligned}$$

6 Unsatisfiability based MaxSat algorithms

Faltaria fer:

- IMPORTANT: Explicar algorismes MSU1, MSU3, **MSU4** i relacionar-los
- IMPORTANT: Explicar algorismes anteriors a weighted i relacionar-los
- Aplicar resolution (row generation) després de column generation
- Identificar els casos on es pot aplicar local search unsat [1] en comptes de systematic: TOTS els casos.
- Com es pot millorar Sat-Unsat: identificant clausules no necessaries (molt difícil)
- Com estendre-ho a weighted, quines techniques es podrien utilitzar

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